RoadGen3D LLM

—

	RoadGen3D main d6f57cf2984ae65 095dd0d75f3c24b7688e7aaeb 2026-07-20 10:10 +08:00
	LLM 2D/3D

"LLM 3D" LLM 3D

- gpt-4o-mini OpenAI Chat Completions
- baseline redesign_default parametric " LLM " auto parametric
- 45/35/20 sidewalk_width_m=3.22 density=0.96 seed=42
- 2D source 3D revision 3D 2D 2D
- structured LLM

1.

LLM

1.
2. LLM
3.
4. LLM

1.1

□□	□□□□	□□□□□
	AOI/OSM/GeoJSON → → source version	3D
	+ + LLM +	LLM
	layout + GLB	LLM
	revision	

1.2

“ _ _ _ ”

2. LLM

2.1

		/
configured	true	API key
provider label	openai	ROADGEN_LLM_PROVIDER
protocol	openai_chat_completions	OpenAI Chat Completions
base URL	https://api.zetatechs.com/v1/	.env GRAPHRAG_API_BASE
endpoint fingerprint	sha256:86 884b773 945…f123f62d	
text model	gpt-4o-mini	ROADGEN_LLM_MODEL/LLM_MODEL
vision model	gpt-4o-mini	text model
temperature	0.2	0.1 0.0
timeout	120 s	
retry	10	429 4 s
max_tokens / top_p / seed		/

“ ”

2.2

ROADGEN_LLM_MODEL ⟶ LLM_MODEL ⟶ gpt-4o-mini ROADGEN_LLM_VISION_MODEL ⟶ Base URL API key ROADGEN_LLM_* GRAPHRAG_* OPENAI_*

C001 C002

3. —

llm_configured “ LLM” “ LLM ” generate("auto")
auto parametric LLM

/			LLM	
baseline	parametric	parametric		/
redesign	auto	parametric		
generation_mode=llm	llm	llm	configured	LLM
	proposal API	side-effect-free proposal		before/after/reason/confidence parametric
LLM llm	llm	parametric fallback		fallback_reason

3.1

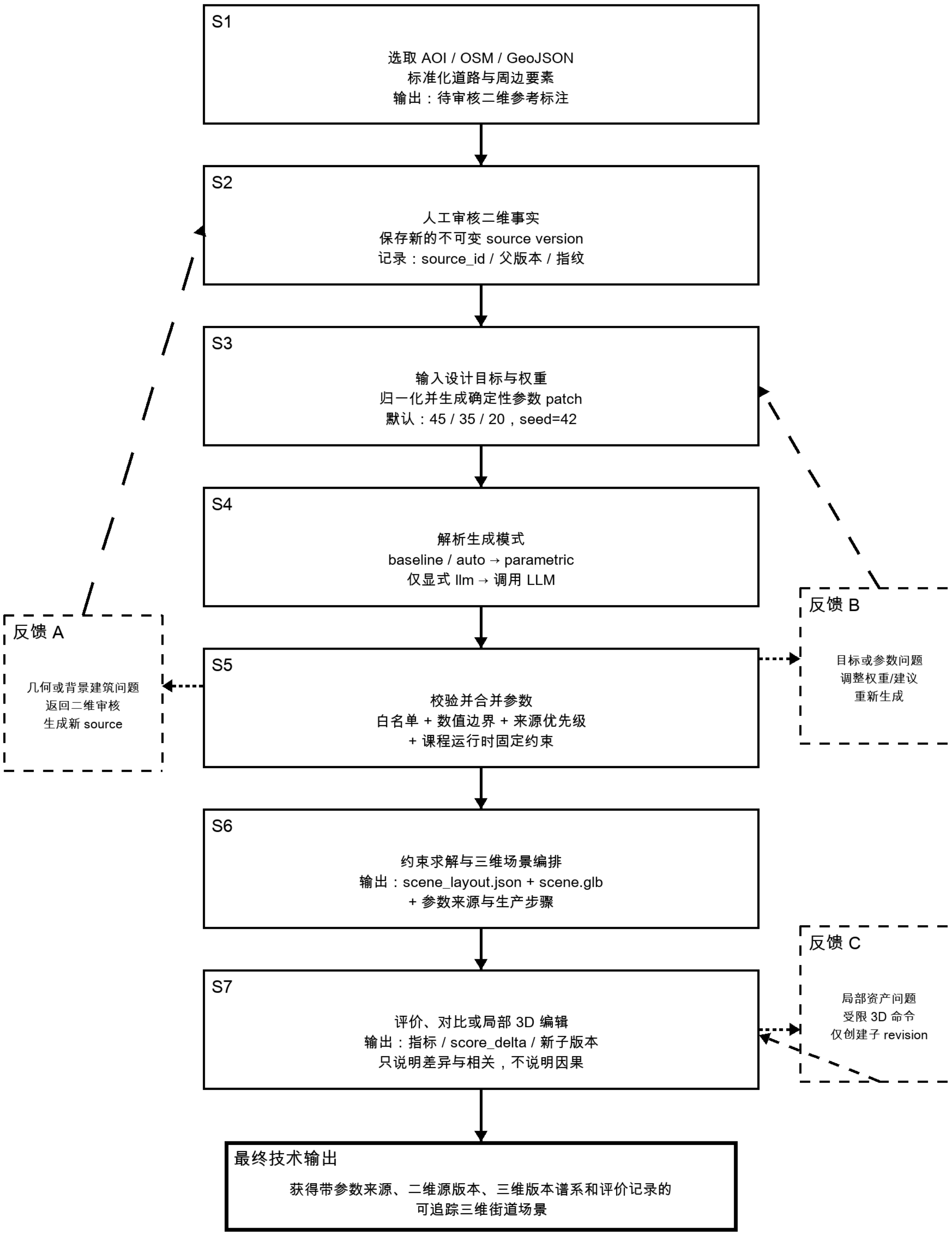
- baseline = parametric redesign_default = parametric parametric_fallback = true
- LLM parameter proposals
- RAG mode = disabled knowledge_source = none

C003 C004 C006 C010 C011

4.

图 1 RoadGen3D 当前生成、修改与反馈主流程

实线：系统数据流 虚线：人工反馈/重新生成路径

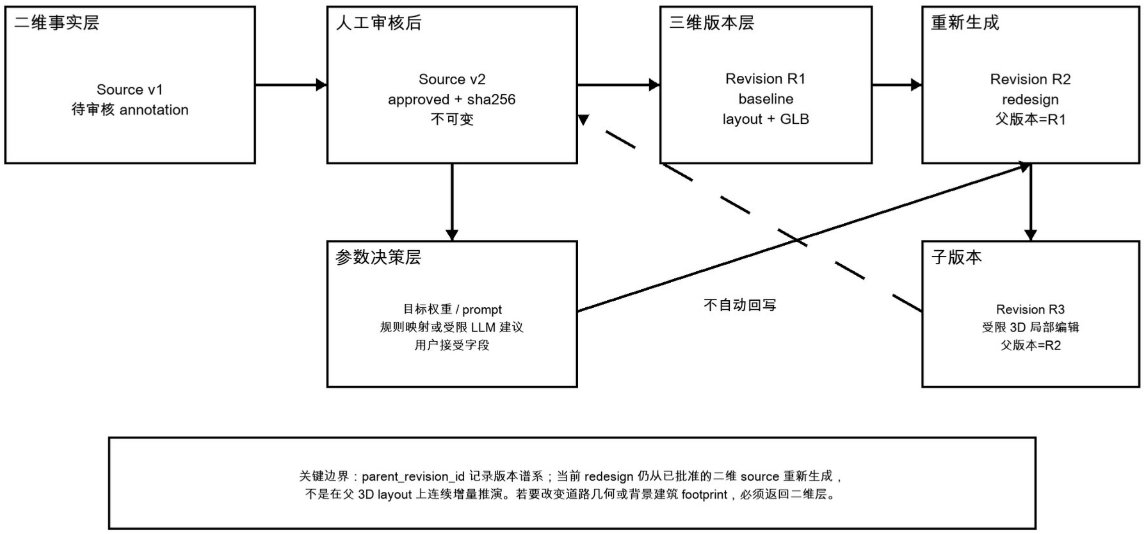


注：当前实现不自动把 3D 修改回写为二维地理事实。

1 “ — — — ” 2D 3D

5.

图 2 二维事实层、参数决策层与三维版本层的交互边界



2 3D 2D redesign parent_revision_id source 3D layout

5.1

- source version parent_source_id review_actions reviewed_at annotation_sha256
- source_id

5.2

- SceneRevisionRecord source_id parent_id branch_kind scene_layout/GLB provenance
- 3D revision OSM/
- footprint 2D source version

6. patch

walkability=45 safety=35 beauty=20 patch

$$w\_i = u\_i / \sum\_j u\_j$$

$$sidewalk\_width\_m = round\left(2.4 + 1.25\,w\_walk + 0.65\,w\_safe + 0.15\,w\_beauty,\ 2\right)$$

$$density = round\left(0.82 + 0.12\,w\_walk + 0.05\,w\_safe + 0.34\,w\_beauty,\ 2\right)$$

	/			
	45 / 35 / 20		0.45 / 0.35 / 0.20	
design_rule_profile	—		pedestrian_priority_v1	
objective_profile	—		balanced	
style_preset	—	+	lush_walkable_v1	
sidewalk_width_m	—	E002	3.22 m	
density	—	E003	0.96	
ped_demand_level	—		high	
bike_demand_level	—	/	medium	
furniture profile	—		pedestrian_friendly	
skeleton profile	—		walkable_commercial	
seed	—		42	

C004 C005 E001–E003

7.

□□□□	□□	□□□□	□□□□□	□□□□□
45/35/20		S3	profile seed	compose patch
	prompt_input / explicit_input	S3–S5		
LLM		S4–S5	skeleton furniture building representation seed	before/after/reason/ confidence
LLM		S5		field source=llm_suggestion parametric
		S5	land-use/ infill knowledge source	
		S6	/ /	scene_layout.json + scene.glb
3D		S7	/ /	revision
	/	S7→S2/S3/S7		score_delta

“ ” source/revision artifact

8. LLM

“AI ” LLM patch /

	LLM	
	laneCount laneWidthM sidewalkWidthM furnishingWidthM curbWidthM junctionCornerPolicy/Radius	geometry
	globalDensity enabled targetCountPer100M spacing setback allowedZones	ID GLB
	representation	building footprint OSM
	seed	
/		GeoJSON approved annotation source version

8.1

	/
laneCount	1–8
laneWidthM	2.5–4.5 m
sidewalkWidthM	1–12 m
furnishingWidthM	0–5 m
curbWidthM	0.05–0.4 m
junctionCornerRadiusM	fixed 1–20 m
globalDensity	0–2
targetCountPer100M	0–20
preferred/minimumSpacingM	2–100 m
roadSetbackM	0–10 m
seed	0–2,147,483,647
	geometryLocked=true footprintLocked=true

9.

patch LLM

60	explicit_input	
57	style blend / transfer	
55	runtime_fixed	/
50	scenario_hard_constraint	
45	prompt_input	
40	parameter_triple	
30	rag_supported_llm	LLM
20	preset_default	
15	llm_derived	LLM
0	default_after_llm	LLM

building_representation=transparent_massing
surrounding_building_mode=footprint_based auto_land_use_mode=off infill_policy=off
building_height_mode=class_only knowledge_source=none retain_glb_policy=always seed=42

C006 C012 C013

10.

		/
move / rotate		revision
scale		0.25–4.0
delete		OSM /
add / duplicate		
replace_asset		provenance
set_building_style		footprint
auto_plant_trees		/ /

10.1 course_grounded

- 0 0–10 m
- planting / furnishing / frontage sidewalk / furnishing / frontage
- viewer manifest editable=false selection_source=osm_white_massing
- scene_layout GLB parent_id

C008 C009 P001

11.

approved source record	source_id parent_source_id annotation_sha256 review_actions	
scene_layout.json		
scene.glb		
SceneRevisionRecord	parent_id branch_kind commands artifact provenance	/
parameter proposal	before after reason confidence sideEffectFree	LLM
structured evaluation		LLM
full visual evaluation	LLM	formal structured composite
revision comparison	score_delta claim_scope	

11.1

- evaluation_mode=structured LLM
- full LLM
- / N/A
- 45/35/20

12.

“ ”

□□□□	□□□□	□□□□□	□□
source + + revision		3D	F001
+ patch		seed	F002
LLM patch + / +	AI		F003
+ patch + runtime_fixed → layout + GLB			F004
3D + +			F005
structured/full			F006

LLM

13.

/		
LLM auto parametric	— LLM	
parent_revision_id 3D layout		redesign
3D 2D		
RAG disabled / knowledge_source=none		
structured LLM		
	/	

13.1

5. “ LLM ” “ — — — ”
6. 45/35/20
7.
8. 3D
9. Zeta /

14.

	/		/
S1–S2	OSM/GeoJSON +	source	source_id parent_source_id annotation_sha256
S3	45/35/20	0.45/0.35/0.20 E002/E003	sidewalk=3.22m density=0.96 profiles seed=42
S4	generation_mode=auto		resolved_generation_mode=parametric
S5	patch +	runtime_fixed	transparent_massing knowledge_source=none
S6	approved annotation + final parameters		scene_layout.json scene.glb production_steps
S7	45/35/20		evaluation score_delta
		/ /	child revision 2D

proposal.reason final parameter source
scene_layout revision provenance

ID	[illegible]	[illegible]	[illegible]	[illegible]
C001	source-code	src/roadgen3d/eval_engine_ext/road_metrics/evaluators/llm_client.py:36-220	LLM / temperature	high
C002	runtime-configuration	.env	API key ROADGEN_LLM_MODEL gpt-4o-mini	high
C003	source-code	web/api/routers/teaching.py:128-145	baseline=parametric redesign_default=parametric RAG disabled LLM	high
C004	source-code	web/viewer/src/react/CourseStudio.tsx:395-490	45/35/20 LLM AI generation_mode=auto	high
C005	source-code	src/roadgen3d/teaching/service.py:128-209	/ profile seed	high
C006	source-code	src/roadgen3d/teaching/service.py:795-1005	2D annotation LLM scene_layout/GLB provenance	high
C007	source-code	src/roadgen3d/teaching/service.py:451-537	2D source version parent_source_id review_actions annotation_sha256	high
C008	source-code	src/roadgen3d/teaching/service.py:1021-1181	3D scene revision GLB/layout artifact AI course_grounded	high
C009	source-code	src/roadgen3d/scene_layout_edits.py:285-475, 556-716	3D revision	high
C010	source-code	src/roadgen3d/services/parameter_proposals.py:16-187	LLM patch geometry footprint GeoJSON GLB ID	high
C011	source-code	web/viewer/src/viewer-parameter-design.ts:55-262	before/after/reason/confidence parametric skip_llm=true	high
C012	source-code	src/roadgen3d/services/street_design_parameters.py:20-326	StreetDesignParameterSpec geometryLocked/footprintLocked profile	high
C013	source-code	src/roadgen3d/services/design_runtime.py:192-275, 590-616, 1069-1354	compose patch LLM LLM	high
C014	source-code	src/roadgen3d/llm/prompts.py:593-700	LLM JSON	high
C015	source-code	src/roadgen3d/llm/design_workflow.py:633-790	structured/full structured LLM LLM formal structured composite	high
C016	source-code	src/roadgen3d/teaching/service.py:1199-1290	evaluation_mode	high
P001	technical-document	docs/teaching-platform.md	2D 3D revision N/A	high

ID	□□	□□	□□□□	□□
P002	technical-document	docs/ ROADGEN3D_3D_GENERATION_TALK.md	/ AI	medium

B

	ReferenceAnnotation approved annotation	
	scene revision revision	
LLM	parameter proposal llm_suggestion	LLM
	parametric skip_llm	LLM
	structured evaluation	
	scene_layout.json	

- explicit F001–F007 explicit
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